Essential Hierarchical Information That Warrants Attention: A Semi-Supervised Hyperspectral Hierarchical Classification Network

Yao Jin, *Graduate Student Member, IEEE*, Yanni Dong, *Senior Member, IEEE*, and Chen Wu, *Member, IEEE*

***Abstract*— The hierarchical classification method can utilize the hierarchical structure to achieve fine classification from coarse-grained to fine-grained. Hierarchical classification has demonstrated outstanding performance across various fields, benefiting from the continuous expansion of dataset sizes. Nevertheless, the majority of current hyperspectral image (HSI) deep learning classification methods employ a flat classification strategy, disregarding the hierarchical structure present in HSI data. Can hierarchical structure information assist deep learning in completing HSI classification tasks? In this article, we validated the feasibility of using hierarchical structure information to assist deep learning in completing HSI classification tasks through intuitive experiments and then proposed a semi-supervised hierarchical classification network (Semi-HCN) to achieve hierarchical classification of HSIs. Semi-HCN first generates a large number of pseudo-labels through a fast self-training (FST) module for subsequent network training and then extracts hierarchical features through a multibranch network and progressively consolidates them across layers. Finally, hierarchical constraints are applied to the network output through hierarchical cross-entropy loss (CE Loss). Experiments on three real datasets have demonstrated that Semi-HCN can achieve competitive classification performance with limited training samples. The code will be released at https://github.com/jinyaoWHU/Semi-HCN**

***Index Terms*— Hierarchical classification, hyperspectral image (HSI), semi-supervised learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) have a high spectral resolution, a wide range of bands, and the characteristic of spatial–spectral integration, which have richer spectral information compared with traditional remote sensing technology. Consequently, HSIs can achieve precise classification of ground objects [1], [2], [3], [4]. Therefore, HSIs are widely used in geological mapping, environmental monitoring, vegetation analysis, atmospheric characterization, biochemical detection, and other fields [5], [6], [7]. HSI classification, as the central problem in the aforementioned scenarios, has garnered significant attention in academic research [8]. Its objective is to assign each pixel in the HSI to a meaningful class.

In the early stages, most classification methods are supervised machine learning methods, such as support vector machines (SVMs) [9], K-nearest neighbors (KNNs) [10], and random forest (RF) [11], which can get satisfactory results under ideal situations. However, these methods heavily depend on expert-designed features, which are usually relatively shallow and difficult to extract deep information from HSIs. This limits the improvement of classification accuracy [12], [13], [14]. With the continuous development of deep learning, it can obtain superior performance, because deep learning can automatically extract more powerful and discriminative features [15].

In recent years, many deep neural networks have been applied in the field of HSI classification, such as stacked autoencoders (SAEs), deep belief networks (DBNs), convolutional neural networks (CNNs), recurrent neural networks (RNNs), and Transformers [16], [17], [18], [19]. Among them, CNNs have attracted a large number of researchers' attention due to their characteristics of local connectivity and weight sharing, which can effectively reduce the number of parameters [20]. In addition, CNNs extensively use convolutional blocks, which enable them to simultaneously capture both spectral and spatial information contained in HSIs. Therefore, more and more CNN-based methods have been proposed and have achieved excellent classification results [21], [22], [23], [24]. The spectral–spatial residual network (SSRN) designs separate spectral residual blocks and spatial residual blocks for extracting rich spectral and spatial features from the input 3-D cube [25]. Hybrid spectral CNN (HybridSN) combines the advantages of 2-D CNNs and 3-D CNNs to further learn the spectral and spatial features in HSIs [26]. As Transformer models demonstrate tremendous potential in various fields related to computer vision and signal processing, numerous Transformer-based classification methods have been proposed. Yu et al. [27] proposed a Multilevel Spectral–Spatial Transformer Network (MSTNet) to address the limitations of existing CNN methods, such as limited receptive fields, information loss in downsampling layers, and high computational resource consumption in deep networks. To address the lack of physical meaning in positional encoding for Transformers in HSI classification, Jiang et al. [28] proposed a novel Graph Generative

Received 21 February 2025; revised 2 April 2025; accepted 23 April 2025. Date of publication 1 May 2025; date of current version 19 May 2025. This work was supported in part by the National Natural Science Foundation of China under Grant T2122014, Grant 62225113, Grant 62222116, and Grant 62171417; and in part by the National Key Research and Development Program of China under Grant 2022YFB3903300. *(Corresponding author: Chen Wu.)*

Yao Jin and Chen Wu are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China (e-mail: yao.jin@whu.edu.cn; chen.wu@whu.edu.cn).

Yanni Dong is with the School of Resource and Environmental Sciences, Wuhan University, Wuhan 430079, China (e-mail: dongyanni@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3566116

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

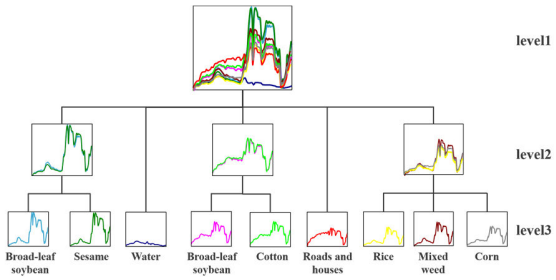


Fig. 1. Tree structure diagram of the WHU-Hi-LongKou dataset.

Structure-Aware Transformer (GraphGST). This approach constructs a new absolute positional encoding to obtain absolute positional sequences of pixels and integrates them into the Transformer architecture, enhancing the understanding of pixel positional relationships.

As the scale and number of data classes increase, the dissimilarity between different classes becomes more apparent, resulting in a natural hierarchical structure within the data. In the field of biology, species can be classified into multiple hierarchical levels, including domain, kingdom, phylum, class, order, family, genus, and species. In urban planning, land-use classes can be organized into a hierarchical structure resembling a tree, based on various types of land utilization. Likewise, HSIs can organize classes into a hierarchical structure resembling a tree, utilizing a specific criterion. As an illustration, the WHU-Hi-LongKou dataset serves as an example to demonstrate the gradual division of classes into multiple hierarchical levels using spectral curve similarity as the criterion. Fig. 1 illustrates the tree structure of the WHU-Hi-LongKou dataset. Based on the hierarchical structure of the data, decision tree methods including ID3 [29], C4.5 [30], and CART [31] have achieved good classification results and have been widely cited. However, the deep learning classification methods mentioned earlier employ a flat classification strategy, which disregards the hierarchical structure of data. Therefore, how to use hierarchical structure information in data has gradually attracted the attention of scholars.

To make full use of this hierarchical structure, scholars have proposed a hierarchical classification strategy. This strategy utilizes the hierarchical structure information within the data as essential auxiliary data to achieve classification from coarse-grained to fine-grained [32]. Generally speaking, hierarchical classification methods can be divided into local classifiers and global classifiers. The local classifier is also known as the top-down classification method [33]. In the prediction process, the local classifier sequentially predicts the classes of the samples, gradually refining the range of selectable classes for subsequent levels of prediction, until the most specific prediction is achieved. Based on the different ways of constructing the classifier, it can be divided into three methods: Local Classifier per Node (LCN), Local Classifier per Level (LCL), and Local Classifier per Parent Node (LCPN) [34]. However, all local classifiers face the problem of error propagation, which means that errors at a certain class level will propagate downward along the hierarchical structure [35]. Furthermore, training multiple groups of classifiers leads to higher time costs. In contrast, the global classifier can directly capture hierarchical information from the data through the network structure or the loss function, resulting in significantly smaller model sizes compared to local classifiers. However, global classifiers make it difficult to achieve local feature mining of hierarchical structures, which is the core feature of local classification methods.

After extensive research, hierarchical classification methods have gained widespread application in various fields, including text classification, art genre classification, disease diagnosis, and other fields [36], [37], [38], [39], [40]. Pereira et al. [41] developed a tree structure to categorize various types of pneumonia and successfully utilized hierarchical classification methods to achieve accurate identification of COVID-19. Chen and Qian [42] proposed a novel loss function based on the hierarchical structure for precise ship classification. Mohammadi and Rustaee [43] implemented a technique involving local classifiers to train a primary classifier and multiple subclassifiers, leading to enhanced artistic style classification capability of the neural network.

Based on the previous discussion, two essential questions deserve consideration. Can hierarchical structure information assist deep learning in completing HSI classification tasks? Can we design a hierarchical classification network specifically tailored for HSI classification tasks to enhance the classification performance of HSIs?

To answer the aforementioned questions, this article conducted experiments on two datasets employing both flat and hierarchical classification strategies to validate the potential of hierarchical structure information. Subsequently, considering HSI classification has long faced the problem of small samples, we adopt a semi-supervised approach to achieve hierarchical classification and propose a new semi-supervised hierarchical classification network (Semi-HCN) for HSI classification. Semi-HCN first generates a large number of pseudo-labels through a fast self-training (FST) module, adds them to the training set, and then completes label fusion based on the hierarchical structure. To extract HSI hierarchical features, we input the original labels and the fused labels into the corresponding branch network for training, and summarize the extracted features layer by layer. Finally, we use hierarchical cross-entropy loss (CE Loss) to impose hierarchical constraints on the network output.

The main contributions of this article are as follows.

1) We explored the expressiveness of flat and hierarchical classification under different training sample conditions. The experimental results demonstrate the potential of hierarchical classification in HSI classification, offering a new solution for further improving classification accuracy.

2) We propose an FST module that does not require model training. This module directly utilizes spatial–spectral information to filter unlabeled samples, efficiently generating a large number of high-precision pseudo-labels, thereby alleviating the limitations imposed by the small-sample problem on classification performance.

3) We propose a hierarchical multibranch network that integrates a multibranch structure with hierarchical information. Each branch corresponds to a specific level of the hierarchy,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

enabling the network to effectively capture the progressive refinement of features at different levels, thereby improving the model's adaptability to hierarchical classification tasks.

4) We propose a hierarchical cross-entropy loss (Hi-CE Loss) that models hierarchical relationships among classes. This loss not only constrains class probabilities but also emphasizes hierarchical relationships. It overcomes the limitations of cross-entropy loss, which fails to incorporate hierarchical constraints.

This article is organized as follows. Section II contains some related works, including self-training, hierarchical classification, and a comparative evaluation experiment of flat and hierarchical classification methods. Section III details the proposed Semi-HCN method. Section IV proves the effectiveness of the Semi-HCN through experiments on three real HSI datasets. Finally, Section V provides some concluding considerations and a vision for future work.

II. Related Works and Preliminary Verification

A. Self-Training

The annotation of HSIs is challenging and costly, often requiring prior knowledge from domain experts and field survey results. Consequently, HSI classification tasks have long faced the issue of limited training samples in practical research. To address this problem, numerous semi-supervised classification methods have been proposed. Su et al. [44] proposed a normalized spectral clustering with kernel-based learning (NSCKL) method based on spectral clustering. NSCKL enhances classification performance by aggregating correlations from local to global scales, thereby obtaining more discriminative embeddings. To address the limitations of source domain methods under large-scale domain shifts, Han et al. [45] proposed a multisource collaborative domain generalization (MSCDG) framework. This framework enhances the model's domain-invariant representation learning capabilities by integrating data-aware adversarial augmentation and model-aware multilevel diversification. It achieves this through the generation of realistic multisource samples, mitigation of interdomain discrepancies, and effective clustering of categories. Pu et al. [46] proposed an end-to-end semi-supervised multimodal dual-path network (SMDN) that integrates HSI and LiDAR data. The network collaboratively trains two types of unsupervised features in a latent subspace to obtain discriminative multimodal representations with a limited number of labeled samples.

Among various semi-supervised methods, self-training approaches have gained widespread application due to their simplicity and ease of implementation, as well as their lack of the need for prior assumptions. In the case of limited labeled samples, the self-training method can use a large number of unlabeled samples for model training, which effectively expands the size of the training set and improves the generalization ability of the model.

For HSIs, we use $D^{\mathrm{l}} = \{(\mathbf{x}_i^{\mathrm{l}}, \mathbf{y}_i)\}_{i=1}^{N}$ to represent the labeled sample set, where $\mathbf{x}_i^{\mathrm{l}} \in R^{1\times C}$ represents the training samples, $\mathbf{y}_i$ represents the corresponding labels, C represents the number of bands, and N is the number of training samples. $D^{\mathrm{u}} = \{\mathbf{x}_i^{\mathrm{u}}\}_{i=1}^{M}$ represents the unlabeled sample set, where $\mathbf{x}_i^{\mathrm{u}} \in R^{1\times C}$ represents the testing samples and M is the number of testing samples. In general, the idea of the self-training method in the HSI classification task is as follows.

First, using label set D^{l} to train an initial classifier θ with CE Loss

$$\min_\theta \sum_{i=1}^{N} L_{\mathrm{CE}}\left(\mathbf{x}_i^{\mathrm{l}}, \mathbf{y}_i\right). \tag{1}$$

Second, using the classifier θ to classify the unlabeled samples, and select the samples with a high confidence score to assign pseudo-labels

$$\mathbf{y}_i^{\mathrm{u}} = f\left(\mathbf{x}_i^{\mathrm{u}}, \theta\right). \tag{2}$$

The generated pseudo-label set $D^p = \{(\mathbf{x}_i^{\mathrm{u}}, \mathbf{y}_i^{\mathrm{u}})\}_{i=1}^{K} \in D^{\mathrm{u}}$ is then combined with the original labeled dataset D^{l}, where K is the number of pseudo-labels. The subsequent round of training is performed on the merged data. Repeat the above steps until the stop condition is reached.

In the process of self-training, the most critical part is how to obtain high-confidence classification results. Because the initial classifier is trained on a small amount of labeled data, and the insufficient number of training samples will lead to insufficient classifier training, it is easy to misjudge the unlabeled data. The errors generated by the unlabeled samples accumulate continuously in the iterative process, which leads to the poor classification effect of the final classifier [47].

Therefore, to improve the performance of self-training methods, many scholars have proposed improved versions of various self-training methods. Transductive SVMs (TSVMs) use dynamic thresholds to select informative samples for SVM classifiers based on self-training methods [48]. Segmentation-based self-learning (SBSL) uses high-resolution image segmentation techniques to select samples with large and diverse information [49]. To effectively train supervised classifiers in self-training, the self-training approach with pseudo (SP-SVM) uses pseudo-label validation sets to improve model classification accuracy and reduce errors [50]. Self-training hierarchy prototype-based (ST-HP) utilized the hierarchical structure existing in the dataset to train a hierarchical classifier in a self-trained manner to improve the semi-supervised classification of remote sensing images [51].

B. Hierarchical Classification

The biggest difference between hierarchical classification and flat classification is that hierarchical classification fully utilizes the hierarchical structure in the data. Hierarchical classification uses hierarchical structure as important side information to achieve top-down classification from coarse to fine granularity.

Silla and Freitas [34] organized existing hierarchical classification methods and proposed a unified hierarchical classification framework. They describe the hierarchical classification method as a 4-tuple $\langle \Delta,\ \Xi,\ \Omega,\ \Theta \rangle$.

1) Parameter Δ: Δ indicates whether the algorithm can assign labels of one or more paths to an instance. That is, whether the algorithm uses multilabel path prediction or

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

single-label path prediction. This parameter offers two possible values: 1) single-path prediction (SSP) indicates that the algorithm can allocate up to one prediction label path for each data instance. 2) Multiple-path prediction (MPP) indicates that the algorithm can allocate up to multiple prediction label paths for each data instance.

2) Parameter Ξ*:* Ξ indicates the prediction depth of the algorithm. This parameter offers two possible values: 1) mandatory leaf-node prediction (MLNP) indicates that the algorithm only assigns class on leaf nodes to an instance. 2) Non-MLNP (NMLNP) indicates that the algorithm can assign classes to an instance on any node.

3) Parameter Ω*:* Ω indicates hierarchical structure type of algorithm processing. This parameter offers two possible values: 1) T (Tree) indicates that the hierarchical structure in the data is a tree structure. 2) D (DAG) indicates that the hierarchical structure in the data is a direct acyclic graph (DAG) structure.

4) Parameter Θ*:* Θ indicates the class of hierarchical classification algorithms. This parameter offers four possible values: 1) LCN training a classifier for each node (except the root node) in the hierarchical structure. 2) LCL training a classifier for each level in the hierarchical structure. 3) LCPN training a classifier for each parent node in the hierarchical structure. 4) GC training a global classifier for all classes. It is worth noting that although the GC model and the flat classification both train a classifier to classify all classes, the GC model will fully consider the hierarchical structure of the data in the training.

C. Experimental Research on Flat Classification and Hierarchical Classification

To verify the feasibility of hierarchical information in HSI classification, we conducted experiments using two classification methods on two different datasets (the Indian Pines dataset and the WHU-Hi-LongKou dataset, which will be introduced in detail in Section IV, including dataset parameter information and hierarchical structure) in this section. By changing the proportion of training samples, we can clearly compare the classification performance of two classification methods. In the experiment, both classification methods used HybridSN as the classifier. The difference is that flat classification only trains one classifier for classification, while hierarchical classification adopts the LCPN, training a classifier on each nonleaf node based on the hierarchical structure. Regarding the setting of training samples, we set the number of training samples to 1, 2, 5, 10, and 20 for each class in two different datasets.

Fig. 2 shows the classification results of two classification methods under different training samples. Overall, the results obtained on two different datasets showed similar trends. When the number of training samples is small, using flattened classification can result in better results. As the number of training samples continues to increase, the accuracy of hierarchical classification increases faster and exceeds the accuracy of flattened classification. This is because a limited number of training samples can result in a high occurrence of misclassifications when employing LCPNs in high-level classification. These misclassifications subsequently propagate in the training of subsequent classifiers, resulting in the utilization of incorrect samples for training. Moreover, inevitably, these erroneous samples will not be accurately classified under this specific classifier.

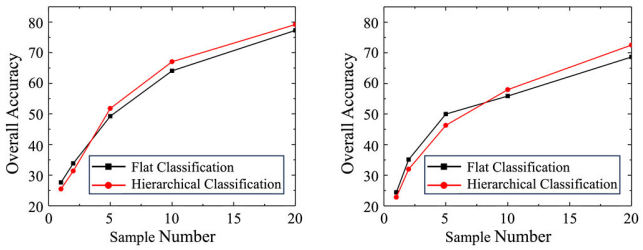


Fig. 2. Comparison of classification results of two different models. (a) Indian Pines. (b) WHU-Hi-LongKou.

III. Proposed Method

In this section, we first introduce the FST module, which is used to generate pseudo-labels for network training. Then, it introduces the multibranch network structure system, which corresponds to the hierarchical structure, and each branch corresponds to the structure of a certain layer of the hierarchical structure. Finally, we introduce the loss function in detail. Algorithm 1 and Fig. 3 illustrate the process of the proposed FST model, while Algorithm 2 and Fig. 4 illustrate the process of the proposed semi HCN.

Algorithm 1 Procedure of the Proposed FST Model

Input: Training set: D^{l}, Testing set: D^{u}, iterations T, Increase training sample P in each iteration

Steps:

for t $= 1 \cdots T$ do

1). Filter Spatial neighbors D^S using 5×5 spatial neighborhood blocks

2). Filter Spatial–Spectral neighbors D^{S-S} using an adaptive threshold b_t based on D^S

3). Obtain the final pseudo-labels through ensemble learning filtering D^p

Update training set $D^{\mathrm{l}} \leftarrow D^{\mathrm{l}} \cup D^{\mathrm{p}}$, $D^{\mathrm{u}} \leftarrow D^{\mathrm{u}} - D^{\mathrm{p}}$

Output: Expanded training set D^{l}

A. FST Model

Self-training is an important means to solve the problem of small samples in hyperspectral classification tasks. However, existing self-training methods train a classifier in each iteration of training and use it to classify unlabeled samples. Multiple iterative training inevitably requires a significant amount of time and computational costs. Therefore, we propose an FST method, which aims to quickly generate a large number of reliable pseudo-labels for subsequent training with a limited number of labeled samples. The FST method uses spatial–spectral information to directly filter unlabeled data, avoiding the classification process in each iteration. This

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

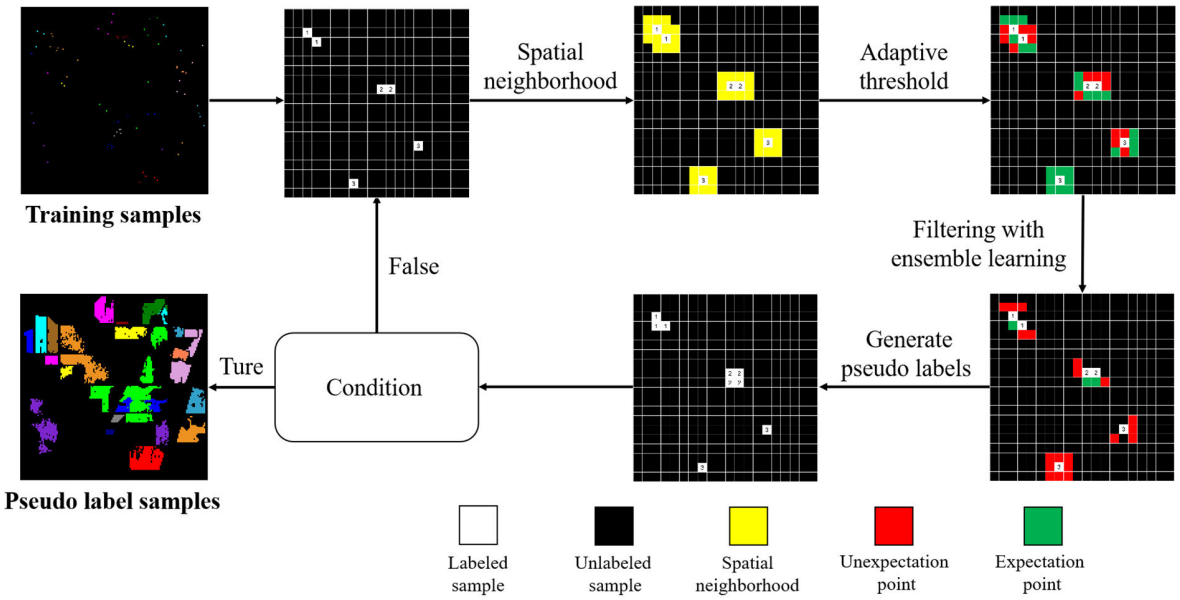


Fig. 3. Architecture of the proposed FST module. Based on the training samples, the unlabeled sample points are first filtered through spatial information to obtain spatial nearest neighbors. Then, an adaptive threshold is used to filter the for the second time to obtain spatial–spectral nearest neighbors. All training samples are collected and filtered for the third time using ensemble learning. Finally, the filtered samples are combined with the initial training sample set and fed into the next iteration of training. After multiple iterations, the expanded training sample set is obtained.

Algorithm 2 Procedure of the Proposed Semi-HCN

Input: Training set: D^1, Hierarchical structure *Tree,*

Steps:

Obtain the expanded training dataset through the FST module

Build hierarchical labels based on the hierarchical structure *Tree*

for *Epochs* $= 1 \cdots 1500$ do

1). Obtain reduced dimensional data using the Data cleaning module

2). The features extracted from the coarse-grained branch are optimized using the CE Loss function, guided by the corresponding hierarchical labels.

3). The features extracted from the coarse-grained branch are aggregated into the fine-grained branch and optimized using the Hi-CE Loss function.

Output: Fine-grained branch classification results

model can ensure the accuracy of pseudo-labels and greatly improve computational efficiency.

1) First Filtering Using Spatial Information: According to the first law of Geoscience, there exists a spatial correlation between features, so the spatial distribution of pixels and features in HSIs is consistent within a certain range. Therefore, adjacent pixels in space are highly likely to represent the same ground object. FST follows this assumption and employs 5×5 spatial neighborhood blocks to select spatial nearest neighbors for labeled samples. The labeled sample is in the center of the block, and the remaining points are the spatial nearest neighbors of the labeled sample points. In this way, each labeled sample can select a maximum of 24 spatial nearest neighbors. However, when the labeled sample is situated at the image's edge, the number of available spatial nearest neighbors may be less than 24.

2) Second Filtering Using Spectral Information: Although the labeled samples are highly similar to the surrounding spatial neighbors, it is impossible to ensure that every spatial neighbor is the same as the labeled sample in the center. Therefore, FST introduces threshold b to further filter the spatial nearest neighbors and screen out the spatial–spectral nearest neighbors. If $d(\mathbf{x}_i, \mathbf{x}_j) > b$, then $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to different classes, and if $d(\mathbf{x}_i, \mathbf{x}_j) < b$, then $\mathbf{x}_i$ and $\mathbf{x}_j$ belong to the same class. $d(\mathbf{x}_i, \mathbf{x}_j)$ is the spectral distance between $\mathbf{x}_i$ and $\mathbf{x}_j$, and its calculation formula is as follows:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \left\|\mathbf{x}_i - \mathbf{x}_j\right\|_2. \tag{3}$$

Considering that the spectral neighborhood information of each labeled sample is not the same, when facing complex data, using a fixed threshold lacks consideration for neighborhood information, which often makes it difficult to achieve good results. So, to fully consider the spectral neighborhood information of each labeled sample and achieve specific problem analysis, this model adopts an adaptive threshold, which calculates the corresponding threshold based on the spectral neighborhood information of each labeled sample and is used to filter the spatial–spectral nearest neighbor points. The specific settings are as follows: for each labeled sample $\mathbf{x}_i$, first, calculate the spectral Euclidean distance between $\mathbf{x}_i$ and its labeled samples of the same class according to (3), obtain the distance matrix between labeled samples of the same class, and take the distance between the labeled samples of the same kind with the nearest t of an as the threshold, and its calculation formula is as follows:

$$b_{it} = \min_t d(\mathbf{x}_i, \mathbf{x}_{ik}),\ \mathbf{x}_i, \mathbf{x}_{ik} \in S \tag{4}$$

where t represents the number of iterations.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

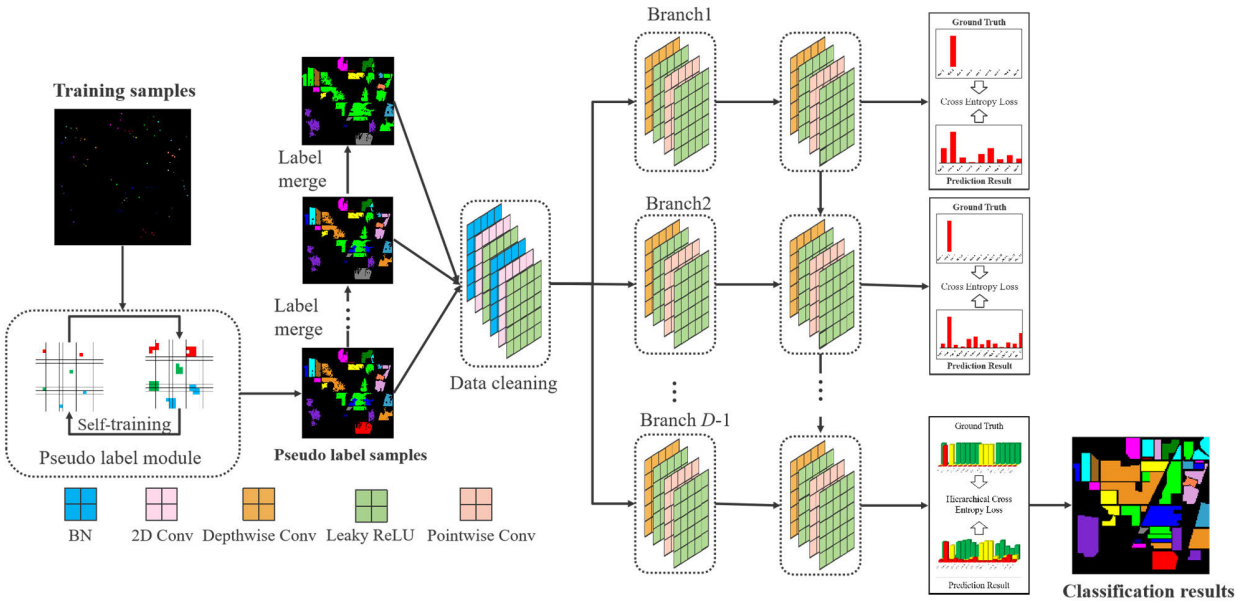


Fig. 4. Architecture of the proposed Semi-HCN. The initial training samples are processed through the FST module to generate pseudo-labels, forming an expanded training set. The expanded training set is then input into the data cleaning module for dimensionality reduction, and the resulting low-dimensional data is fed into different branches for processing. In the coarse-grained branch, the corresponding hierarchical labels are used to compute the CE Loss. The features extracted by the coarse-grained branch are progressively aggregated into the fine-grained branch, where they are further constrained and optimized using the Hi-CE Loss. Finally, the output from the fine-grained branch is taken as the final classification result.

3) Final Filtering Using Ensemble Learning: Once the unlabeled samples undergo two rounds of filtering based on spatial–spectral information, each sample point obtains its respective set of spatial–spectral nearest neighbors. To further enhance the accuracy of pseudo-labels, FST incorporates the concept of ensemble learning to filter the spatial–spectral nearest neighbors for the last time. Pseudo-labels are assigned to each unlabeled sample only if it is identified as a spatial–spectral nearest neighbor point by two or more labeled samples.

B. Network Structure

1) Data Cleaning Modules: To solve the problem that the original HSI contains noise and redundant information, which affects the network training. Inspired by the network in the network [52], we used two 1×1 convolution layers as data cleaning modules in the initial stage of the network. Their primary purpose is to facilitate cross-channel information exchange, eliminating redundant spectral information and enhancing recognition capabilities. In addition, the data-cleaning module can also be used as a dimension reduction module to reduce the dimension of the original data, effectively avoiding the Hughes phenomenon that is prone to occur in HSI classification.

2) Feature Extraction Module: Considering that the FST module will generate a substantial number of pseudo-label samples for training, our network utilizes deep separable convolution to construct a 2DCNN for extracting contextual spatial features. This approach effectively reduces parameters and runtime, while improving robustness and mitigating the risk of overfitting during training.

3) Multibranch Structure: To effectively leverage the hierarchical structure of the data, our network structure employs a multibranch design, with each branch corresponding to a specific layer of the tree hierarchy. Different branches adopt the same feature extraction structure design. The distinguishing factor lies in the disparity between the input label information and the utilized loss function. Specifically, there are two types of branches: 1) coarse-grained branch and 2) fine-grained branch (in the network structure, all branches except the final branch are coarse-grained branches, while the last branch is fine-grained). Based on the coarse-grained branches, the label information will be fused according to the structure of the child node before entering the label information. For example, if C1 and C8 in the Indian dataset belong to the same parent node, the two classes will be regarded as the same class when entering the coarse-grained branches. After feature extraction, the traditional CE Loss will be used for training. At the same time, the features extracted from each coarse-grained branch will be input into the next coarse-grained branch, and finally collected into the fine-grained branch. The input of the fine-grained branch is the original label information. After feature extraction, it is fused with the features extracted from multilayer coarse-grained branches and trained using our proposed Hi-CE Loss.

C. Loss Function

1) Cross-Entropy Loss: In the coarse-grained branch, we use CE Loss as the loss function, which is the most common loss function in classification tasks. The CE Loss function can effectively measure the difference between predicted results and actual results, and its specific calculation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

formula is as follows:

$$\text{CE Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{\text{C}} p_{ij}\log\left(q_{ij}\right) \tag{5}$$

where q_{ij} represents the probability of network output, p_{ij} represents the true label, C represents the number of classes, and N represents the number of samples.

2) Hierarchical Cross-Entropy Loss: The traditional CE Loss emphasizes the importance of correct classes by maximizing the probability of correct classes in the network output. However, CE Loss only emphasizes the correct classes and cannot further constrain other classes. When facing hierarchical classification tasks, CE Loss cannot use the hierarchical structure to impose more constraints on the network data so that the hierarchical structure of the network output remains consistent in the data. Therefore, we propose a Hi-CE Loss for a fine-grained branch based on the traditional CE Loss, aiming to impose hierarchical constraints on the network output, ensuring the network output aligns with the hierarchical structure of the data and enhancing the network's classification capability through the utilization of hierarchical structure. The specific calculation formula is as follows:

$$\text{Hi-CE Loss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{\text{C}}\sum_{k=1}^{D}\hat{\mathbf{p}}_{ijk}\log\left(\hat{\mathbf{q}}_{ijk}\right) \tag{6}$$

where D is the depth of the hierarchical structure. Fig. 5 illustrates the difference between CE Loss and Hi-CE Loss using the same sample point $\mathbf{x}_i$ as an example, which class is C5. The biggest difference lies in the fact that, in the case of CE Loss, the true one-hot label p_i corresponding to a sample point is a vector of size $C \times 1$,whereas, in Hi-CE Loss,we expand the $C \times 1$ vector p_i into a matrix $\hat{\mathbf{p}}_i$, which size is $C \times D$. The $\hat{\mathbf{p}}_{ij}$ reflects the relationship between Class j and the correct class in the hierarchical structure. If Class j and the correct class belong to the same parent node at the d level, then $\hat{\mathbf{p}}_{ijd} = 1$, and the rest of the values in that column are 0. For example, if C3 and C5 belong to the same parent node in the level 2, then $\hat{\mathbf{p}}_{i32} = 1$. And for the correct class, we set $\hat{p}_{ijD} = 1$, for $\mathbf{x}_i$, $\hat{\mathbf{p}}_{i5D} = 1$. By constructing the ground-truth labels in this way, we can make the network's output conform to the hierarchical structure of the data. We can observe that $\hat{\mathbf{p}}_{ij}$ is the same as p_i in the CE Loss, so we can apply Softmax layer to $\hat{\mathbf{p}}_{ij}$ to obtain the final classification result. Additionally, to mitigate excessive constraints imposed on the hierarchical structure and the potential dilution of constraints on the accurate class, Hi-CE Loss integrates the usage of p_i to calculate CE Loss, thereby highlighting the significance of the correct class. Consequently, the definitive computational formula for Hi-CE Loss is presented as follows:

$$\begin{aligned}\text{Hi-CE Loss} &= -\frac{\alpha}{N}\sum_{i=1}^{N}\sum_{j=1}^{\text{C}}\sum_{k=1}^{D}\hat{\mathbf{p}}_{ijk}\log\left(\hat{\mathbf{q}}_{ijk}\right) \\ &\quad + \left(-\frac{1-\alpha}{N}\sum_{i=1}^{N}\sum_{j=1}^{\text{C}}\sum_{k=D-1}^{D}\hat{\mathbf{p}}_{ijk}\log\left(\hat{\mathbf{q}}_{ijk}\right)\right).\end{aligned} \tag{7}$$

TABLE I
LAND-COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF THE INDIAN PINES DATASET

No	Class	Train	Val	Test
C1	Alfalfa	2	2	42
C2	Corn-notill	2	2	1424
C3	Corn-min	2	2	823
C4	Corn	2	2	233
C5	Grass-pasture	2	2	479
C6	Grass-tree	2	2	726
C7	Grass-pasture-mowed	2	2	24
C8	Hay-windrowed	2	2	474
C9	Oats	2	2	16
C10	Soybeans-notill	2	2	968
C11	Soybeans-min	2	2	2451
C12	Soybeans-clean	2	2	589
C13	Wheat	2	2	201
C14	Woods	2	2	1261
C15	Bldg-grass-tree-drives	2	2	382
C16	Stone-stell-towers	2	2	89
	Total	32	32	10182

IV. EXPERIMENTS

This section will introduce the dataset and experimental setup used in our experiment, summarize the experimental results, and briefly analyze the proposed Semi-HCN.

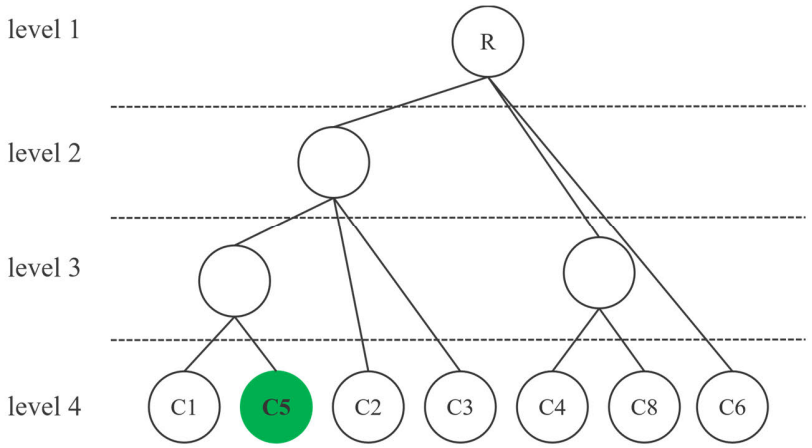
A. Dataset Description

To comprehensively evaluate the performance of the proposed algorithm, the experiment was conducted on three sets of real hyperspectral datasets, including the Indian Pines dataset, the Pavia University dataset, and the WHU-Hi-LongKou dataset. Currently, these three datasets lack a clearly defined hierarchical structure. Therefore, utilizing the pseudo-labels generated by the FST module, we calculate the interclass distance between different classes and subsequently construct a hierarchical structure based on this distance.

1) Indian Pines Dataset: The Indian Pines dataset [53], which was collected in 1992, is one of the earliest datasets used for HSI classification. It was captured by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor and consists of an image with a spatial resolution of 20 × 20 m, covering 145 × 145 pixels. The dataset encompasses a wavelength range of 0.4–2.5 μm and comprises 220 continuous bands. After excluding 20 bands affected by water absorption and noise, 200 bands are retained. Of the total 21 025 pixels, approximately half (10 249 pixels) are labeled into 16 distinct classes. For further information regarding the classes and division of the dataset, refer to Table I. Fig. 6(a)–(c) displays the corresponding false-color image ground-truth map, ground-truth map, and hierarchical structure, respectively.

2) Pavia University Dataset: The Pavia University dataset was captured using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor in 2001 at the University of Pavia, Italy [54]. It has a size of 610 × 340, consisting

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.



One-hot label for CE Loss

C1	C2	C3	C4	C5	C6	C7
0	0	0	0	1	0	0

One-hot label for Hi-CE Loss

C1	C2	C3	C4	C5	C6	C7
0	0	0	1	0	1	1
0	1	1	0	0	0	0
1	0	0	0	0	0	0
0	0	0	0	1	0	0

Fig. 5. Example of Hi-CE Loss. We use one-hot labels to illustrate the differences between the two loss functions. In our proposed Hi-CE Loss, the 1-D labels in the standard CE Loss are extended to a 2-D format. This extension not only constrains the correct class but also enforces the relationships between classes. Each column in the label matrix represents the relationship between a given class and the correct class. For instance, assuming C5 is the correct class, if C1 and C5 share the same parent node at level 3, the third row of the column corresponding to C1 is assigned a value of 1.

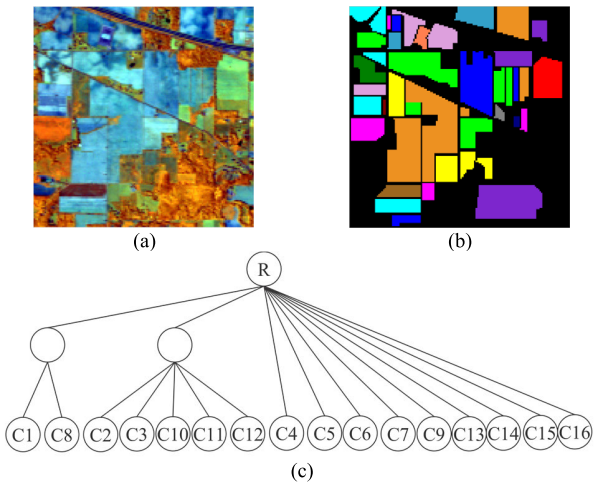


Fig. 6. Indian Pines dataset. (a) False-color image. (b) Ground-truth map. (c) Hierarchical structure.

of a total of 207 400 pixels. The spatial resolution of the dataset is 1.3 m, and the wavelength range spans from 0.43 to 0.86 μm, encompassing 115 continuous bands. After eliminating the bands that cause interference, the remaining 103 bands are utilized for research purposes. The ground-truth information available for this dataset includes approximately 20% of the pixels (42 776 out of 207 400), which are labeled into nine different classes. For more detailed information regarding each class and the division of the dataset, refer to Table II. Fig. 7(a)–(c) displays the corresponding false-color image ground-truth map, ground-truth map, and hierarchical structure, respectively.

3) WHU-Hi-LongKou Dataset: The WHU-Hi-LongKou dataset was obtained in LongKou City, China, in 2017, using an unmanned aerial vehicle (UAV) platform [55]. The dataset has a size of 550 × 400 pixels, encompassing a total of 220 000 pixels. It has a resolution of 0.463 m and covers a wavelength range from 0.4 to 1 μm, consisting of 270 continuous bands. The ground-truth information available for this dataset covers approximately 90% of the pixels (204 542 out of 220 000), which are labeled into nine different classes. For more detailed information regarding each class and the division of the dataset, refer to Table III. Fig. 8(a)–(c) displays the corresponding false-color image ground-truth map, ground-truth map, and hierarchical structure, respectively.

TABLE II

LAND-COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF THE PAVIA UNIVERSITY DATASET

No	Class	Train	Val	Test
C1	Asphalt	2	2	6627
C2	Meadows	2	2	18645
C3	Gravel	2	2	2095
C4	Trees	2	2	3060
C5	Metal Sheets	2	2	1341
C6	Bare Soil	2	2	5025
C7	Bitumen	2	2	1326
C8	Bricks	2	2	3678
C9	Shadow	2	2	943
	Total	18	18	42740

TABLE III

LAND-COVER CATEGORIES AND THE DATASET DIVISION FOR EACH CATEGORY OF THE WHU-HI-LONGKOU DATASET

No	Class	Train	Val	Test
C1	Corn	2	2	34507
C2	Cotton	2	2	8370
C3	Sesame	2	2	3027
C4	Broad-leaf soybean	2	2	63208
C5	Narrow-leaf soybean	2	2	4147
C6	Rice	2	2	11850
C7	Water	2	2	67052
C8	Roads and houses	2	2	7120
C9	Mixed weed	2	2	5225
	Total	18	18	204506

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

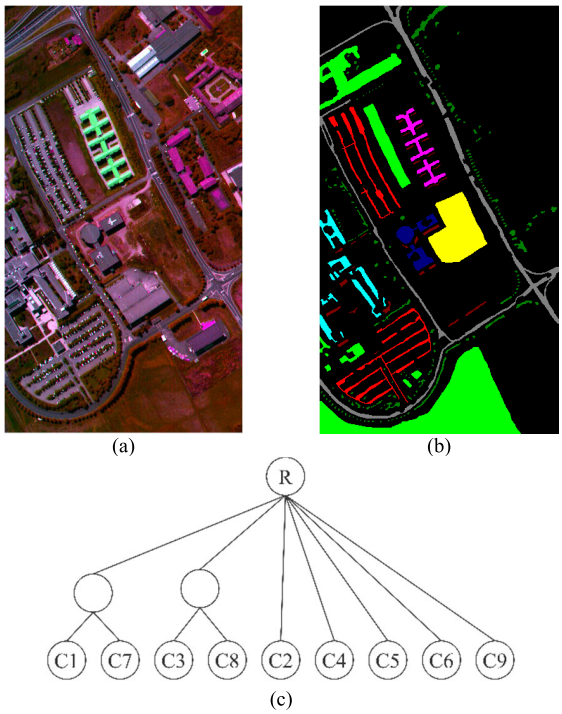


Fig. 7. Pavia University dataset. (a) False-color image. (b) Ground-truth map. (c) Hierarchical structure.

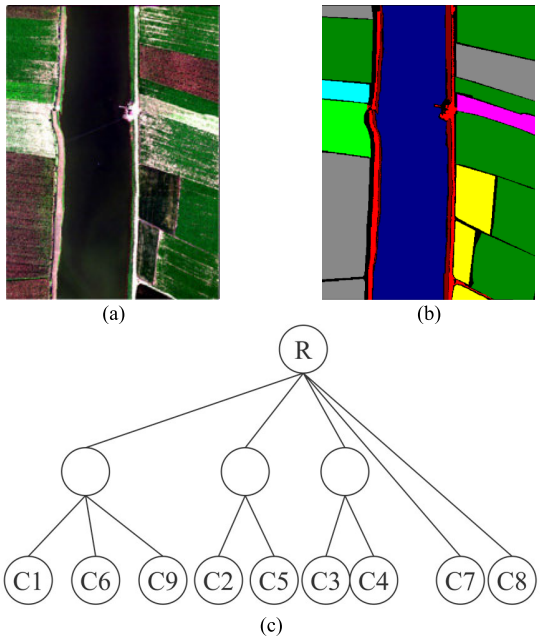


Fig. 8. WHU-Hi-LongKou dataset. (a) False-color image. (b) Ground-truth map. (c) Hierarchical structure.

B. Experimental Setting

1) Implementation Details: Our model is implemented based on Python 3.10.12 and Python 1.7.0. The i7-10700K CPU and NVIDIA GeForce RTX 3090 GPU were used as experimental hardware. In the experiments, the same parameter settings are used on different datasets. We use the Adam optimizer to train the model with a learning rate of 0.0001, and the number of epochs is 1500. There are two parameters maximum iterations T and hyperparameter α need to be manually set in our algorithm. Considering the running time, we set the maximum iterations $T = 50$, and based on the parameter analysis results, we set $\alpha = 0.9$. For partitioning the training set, validation set, and testing set, we select two labeled samples in each class as the training and validation set, while assigning the remaining samples as test data. Moreover, to ensure a fair comparison, all methods employed the same ground-truth data throughout the experimental process. To assess the classification performance of our algorithm both quantitatively and qualitatively, we employ the following evaluation criteria: overall accuracy (OA), average accuracy (AA), kappa coefficient, and per-class accuracy (CA). Furthermore, each experiment was repeated ten times, and the final result was the average of the ten experiments to reduce errors caused by random sampling.

2) Comparison Methods: To effectively validate the performance of the proposed algorithm, seven state-of-the-art algorithms were selected as comparative algorithms in the experimental section, including two supervised algorithms, that is, the HybridSN and attention-based adaptive spectral–spatial kernel improved residual network (A^2S^2K-ResNet) [56], two self-supervised algorithms, that is, the spectral–spatial Siamese network (S^3Net) [57] and multistage relation network with dual metric (DM-MRN) [58], and three semi-supervised algorithms, that is, the CNN-Enhanced Graph Convolutional Network (CEGCN) [59], End-to-end Mixhop Superpixel-based GCN (EMS-GCN) [60], Weighted Feature Fusion of CNN and Graph Attention Network (WFCG) [61], Decision Tree (DT) [62], and RF. In addition, to compare the ability of Semi-HCN and other hierarchical classification methods in utilizing hierarchical structure, we set the RF model integrated with the FST module (FST-RF) as the comparison method. All algorithms are run in the same environment and use default values from open-source code or hyperparameters recommended in the original article to ensure fairness in the comparative experiment. The detailed parameter settings for the comparison algorithms are as follows.

1) *HybridSN and A^2S^2 KResNet:* The experimental settings include a patch size of 9, batch size of 32, and learning rate of 0.0005.
2) *S^3 Net:* The experimental settings include a patch size of 9, batch size of 32, learning rate of 0.001, and PCA reduced to 60 components.
3) *DM-MRN:* PCA is reduced to 30 components, and the top-k is set to 5.
4) *EMSGCN, CEGCN, and WFCG:* The segmentation scale parameter (λ) is set to 100, and the learning rate is 0.001.
5) *DT:* There are no parameters that need to be manually set.
6) *RF:* The number of trees in the forest is set to 100.
7) *FST-RF:* The number of trees in the forest is set to 100, and the maximum iterations in FST is set to 50.

C. Performance on the Three Datasets

In this section, we use three real-world datasets to quantitatively and qualitatively evaluate various classification methods to demonstrate the effectiveness of our approach.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASSIFICATION RESULTS OF THE INDIAN PINES DATASET

Class	HybridN	A^2S^2KResNet	S^3NET	DM-MRN	CEGCN	EMSGCN	WFCG	DT	RF	FST-RF	Ours
C1	33.96	16.90	94.54	23.49	63.09	95.95	87.14	30.95	92.85	97.61	**100**
C2	27.80	42.34	41.03	28.14	39.45	39.00	39.03	23.73	22.75	54.49	**76.12**
C3	19.94	25.59	50.78	22.58	40.63	38.80	36.17	16.94	16.34	40.67	**55.32**
C4	2.88	30.48	66.17	27.28	66.95	71.67	84.46	30.04	20.17	93.56	**100**
C5	25.64	71.40	66.13	**92.92**	59.66	58.16	57.93	16.70	70.77	82.88	65.55
C6	19.32	83.33	93.13	82.01	64.31	66.83	77.17	39.53	17.76	71.62	**93.66**
C7	1.80	31.33	**100**	12.87	**100**	99.16	99.16	54.16	91.66	**100**	95.83
C8	20.12	94.60	61.89	94.43	93.58	97.17	98.71	56.54	19.83	47.89	**99.36**
C9	1.88	5.01	97.77	5.24	90.62	88.75	**100**	43.75	43.75	**100**	**100**
C10	17.48	36.12	55.26	37.02	63.95	69.61	67.01	39.56	66.63	68.69	**75.41**
C11	43.26	54.57	45.83	**84.20**	58.00	52.57	52.50	10.28	20.19	36.51	53.48
C12	21.93	30.36	46.64	32.54	48.86	49.47	65.84	22.41	18.16	49.74	**74.70**
C13	8.07	74.66	98.91	49.13	96.21	99.50	98.55	66.16	89.05	39.80	**100**
C14	46.42	**89.72**	82.88	89.21	76.14	82.40	84.01	52.26	80.17	77.87	70.34
C15	19.84	61.66	65.88	34.08	76.93	75.41	79.63	39.26	9.94	97.64	**99.73**
C16	25.00	69.10	**99.78**	19.83	95.05	98.98	90.33	87.64	80.89	100	98.87
OA (%)	24.07 ±6.33	45.97 ±4.05	59.35 ±4.49	55.05 ±3.74	60.63 ±7.68	61.02 ±7.59	62.82 ±5.99	29.48 ±3.38	36.16 ±4.91	58.24 ±2.97	**72.24 ±3.99**
AA (%)	14.94 ±6.72	40.92 ±3.23	72.91 ±4.65	45.94 ±1.39	70.84 ±4.37	73.96 ±4.05	76.10 ±2.58	39.37 ±2.85	47.56 ±2.62	72.44 ±1.31	**84.90 ±1.53**
kappa×100	20.96 ±5.60	51.07 ±4.32	54.68 ±1.43	50.14 ±4.05	56.03 ±7.96	56.65 ±8.10	58.45 ±6.20	23.69 ±3.36	29.67 ±4.79	53.68 ±2.92	**68.84 ±4.43**
Time (s)	18.15	65.19	101.36	401.5	15.23	28.10	23.88	0.09	0.42	12.77	30.58

TABLE V
CLASSIFICATION RESULTS OF THE PAVIA UNIVERSITY DATASET

Class	HybridSN	A^2S^2KResNet	S^3NET	DM-MRN	CEGCN	EMSGCN	WFCG	DT	RF	FST-RF	Ours
C1	40.03	86.86	69.97	76.20	60.92	63.28	72.61	48.93	53.02	66.89	**88.47**
C2	76.76	41.37	56.69	**99.35**	64.57	60.41	57.61	16.98	41.93	64.52	72.13
C3	15.69	51.54	60.53	44.00	**85.10**	78.30	79.11	31.07	53.55	48.25	58.98
C4	52.35	**92.20**	82.57	41.21	65.27	85.20	80.21	89.64	84.57	80.26	79.07
C5	32.10	33.09	99.87	88.88	99.70	99.41	99.80	97.61	71.81	99.55	**99.99**
C6	20.21	45.13	75.81	48.74	88.57	83.23	**89.90**	50.76	56.57	84.53	87.06
C7	6.31	63.24	**98.81**	50.64	98.69	98.16	96.25	62.74	90.72	69.53	83.27
C8	21.22	63.24	85.07	65.48	77.68	83.11	83.22	87.62	47.58	81.26	**90.14**
C9	36.78	69.32	96.40	80.08	72.06	84.11	87.09	99.15	**100**	99.78	99.35
OA (%)	33.98 ±7.51	59.93 ±5.72	69.02 ±6.74	65.03 ±4.16	71.33 ±8.05	71.06 ±7.49	71.78 ±6.81	43.65 ±9.65	53.22 ±7.84	71.05 ±11.22	**79.64 ±4.92**
AA (%)	21.69 ±9.92	51.69 ±5.82	80.63 ±3.27	66.06 ±4.46	79.17 ±4.23	81.69 ±5.27	82.8 7±3.43	64.95 ±8.70	66.64 ±4.54	77.18 ±3.65	**84.27 ±3.31**
kappa×100	33.50 ±5.84	63.38 ±3.51	62.23 ±6.94	58.11 ±3.36	64.70 ±8.81	64.56 ±8.28	65.65 ±7.40	34.60 ±9.48	43.74 ±7.45	63.83 ±11.70	**74.21 ±5.72**
Time (s)	28.18	105.47	224.93	417.17	50.89	109.36	209.55	0.06	0.37	10.88	64.71

TABLE VI
CLASSIFICATION RESULTS OF THE WHU-HI-LONGKOU DATASET

Class	HybridSN	A^2S^2KResNet	S^3NET	DM-MRN	CEGCN	EMSGCN	WFCG	DT	RF	FST-RF	Ours
C1	83.28	87.96	92.32	94.03	93.57	96.75	93.79	89.83	55.07	84.23	**99.90**
C2	36.35	65.12	81.30	69.54	74.81	83.23	71.71	29.12	21.26	89.88	**93.62**
C3	7.18	48.02	93.14	60.07	**97.26**	94.39	95.44	66.50	73.93	76.87	87.04
C4	91.05	93.23	70.88	**97.58**	76.34	69.6	80.96	46.43	62.99	70.69	84.98
C5	22.29	39.92	**96.19**	56.52	94.03	93.34	91.68	43.95	65.34	32.07	94.67
C6	61.24	82.41	78.77	66.20	82.59	85.46	87.68	62.18	68.35	77.05	**88.18**
C7	99.34	97.31	96.89	98.16	97.55	99.71	91.90	**99.96**	99.10	99.82	99.95
C8	**71.41**	69.67	61.39	61.87	56.97	59.70	63.06	54.74	34.62	29.56	61.08
C9	36.24	31.56	67.59	52.86	60.44	55.94	**71.78**	2.47	54.50	46.41	63.67
OA (%)	77.31 ±1.60	82.31 ±6.20	84.34 ±4.44	84.15 ±3.55	86.09 ±2.82	85.78 ±5.20	86.30 ±4.67	70.93 ±6.24	71.40 ±9.45	80.94 ±7.24	**91.80 ±3.05**
AA (%)	71.06 ±2.00	77.49 ±7.39	82.06 ±2.54	72.98 ±4.42	81.50 ±2.66	82.05 ±3.59	83.11 ±4.03	55.03 ±5.48	59.57 ±7.27	75.65 ±7.65	**85.90 ±4.23**
kappa×100	56.49 ±1.73	68.36 ±6.56	80.06 ±5.35	79.93 ±4.28	82.25 ±3.36	81.99 ±6.15	82.61 ±5.50	63.63 ±6.86	64.27 ±10.42	67.40 ±5.43	**89.38 ±3.81**
Time (s)	45.34	104.56	256.34	635.49	60.53	124.63	240.06	0.03	0.38	149.40	164.35

1) Quantitative Evaluation: Tables IV–VI show the mean and variance of experimental results obtained using different classification methods on three datasets, including OA, AA, kappa, and CA. In the tables, bold numbers represent the highest value.

It can be seen from the experimental results that the experimental results obtained by using the semi-supervised classification method are much better than those obtained by using the supervised classification method in the case of a few training samples. This is because only using the information

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
NUMBER OF PSEUDO-LABELS, ACCURACY, COMPUTATION TIME, AND THE CORRESPONDING OA UNDER DIFFERENT POLICY COLUMN COMBINATIONS ON THREE DATASETS

	Indian Pines				Pavia University				WHU-Hi-LongKou			
	OA(%)	Number	Acc(%)	Time(s)	OA(%)	Number	Acc(%)	Time(s)	OA(%)	Number	Acc(%)	Time(s)
A	60.02	5047	84.78	1.53	81.09	5202	100	3.00	89.90	10062	95.60	4.30
A+B	65.67	5026	89.98	9.60	80.34	5123	100	8.30	91.22	10589	99.94	28.82
A+C	64.88	5065	90.48	6.53	80.19	5063	100	3.75	90.41	10292	98.42	5.61
A+B+C	69.36	5048	93.09	25.93	80.21	5027	100	18.69	91.40	10058	99.97	57.51

TABLE VIII
OA (%) INDICES OF THE COMBINATIONS OF DIFFERENT MODULES ON THREE DATASETS

FST	Multi branch	Hierarchical labels	Hi-CE Loss	Indian Pines	Pavia University	WHU-Hi-LongKou
				47.96	60.56	77.58
	√			49.54	61.14	80.62
		√		49.79	62.06	81.74
			√	50.96	61.75	79.68
		√	√	51.87	63.43	83.21
√				69.47	76.35	89.44
√	√			69.58	77.51	90.22
√		√		70.69	78.09	90.69
√			√	69.97	78.44	90.45
√		√	√	72.32	78.96	92.07

contained in the limited training samples is not enough to train a good classifier. This also proves that the supervised learning method makes it difficult to deal with the small sample problem in the HSI classification task. In the semi-supervised classification method, the superpixel-based GCN can automatically learn the high-level representation of superpixels and has better classification results. Both CEGCN and WFCG integrate features from the CNN and GCN. However, due to the CEGCN direct concatenation approach for feature fusion, it encounters difficulties in merging features. On the other hand, WFCG can better explore the complementarity between CNN and GCN features, which leads to superior classification results. Compared with FST-RF, our method has better classification accuracy using FST with the same settings, indicating that Semi-HCN can better extract hierarchical information from data to assist in classification tasks.

On the whole, our proposed method achieves the highest OA, AA, and kappa on the three sets of datasets. Specifically, compared with the suboptimal method, our method improves the classification accuracy by 9.42% for OA, 8.80% for AA, 10.35% for kappa coefficient on the Indian Pines dataset, 7.86% for OA, 1.40% for AA, 8.56% for kappa coefficient on the Pavia University dataset, and 5.50% for OA and 2.79% for AA, 6.77% for kappa coefficient on the WHU-Hi-LongKou dataset, which proves the effectiveness of our method for HSI classification. From the results of CA, the class of internal nodes in the tree structure have better classification accuracy, such as C2 (cotton) and C5 (broad-leaf soybean) in the WHU-Hi-LongKou dataset, which has the highest or second highest classification accuracy.

2) Qualitative Evaluation: To demonstrate the classification results more intuitively, Figs. 9–11 show the land-cover classification maps obtained from different methods on three datasets. From the figures, it can be observed that HybridSN and A^2S^2K-ResNet exhibit noticeable misclassification issues, with blurred land-cover boundaries. This phenomenon is primarily attributed to the insufficient number of training samples, which hinders the extraction of adequate spatial features. DM-MRN utilizes the powerful feature learning ability of relationship networks in the case of small samples, achieving intraclass cohesion and interclass separation, resulting in low misclassification rates for classification. Although the superpixel-based semi-supervised graph learning method reduces misclassification occurrences, it still suffers from pronounced oversmoothing, which results in blurred boundaries between land-cover classes. In comparison, DT, RF, and FST-RF methods present clearer delineation of land-cover boundaries, but they also exhibit significant misclassification issues. Compared to other models, Semi-HCN provides sufficient training samples for network training through the use of pseudo-labels, yielding the best classification results. Moreover, Semi-HCN fully exploits the inherent hierarchical structure of HSI data, which not only enables the precise preservation of fine edge details but also facilitates more effective differentiation of easily confused classes, such as C2 (corn-notill), C3 (corn-min), C10 (soybeans-notill), C11 (soybeans-min), and C12 (soybeans-clean) in the Indian Pines dataset, C1 (corn), C6 (rice), and C9 (mixed weed) in the WHU-Hi-LongKou dataset.

D. Discussion

1) Ablation Experiment: This section focuses on showing a series of ablation experiments to explore how the different modules contribute to the performance gain. Table VII illustrates the impact of different strategies in the FST module on pseudo-label generation. The FST module primarily employs two strategies: adaptive thresholding and ensemble learning-based filtering. The baseline of FST only utilizes spatial–spectral information to complete the first two filtering

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

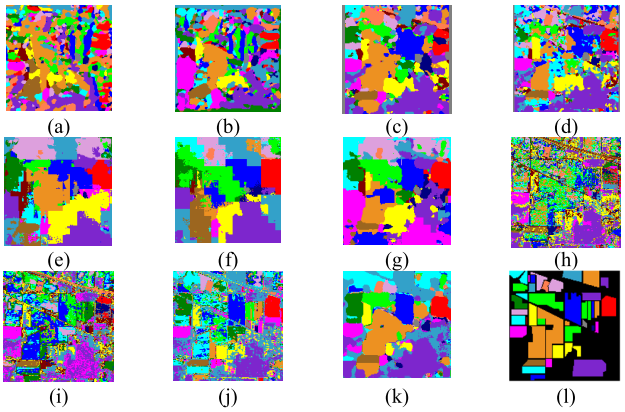


Fig. 9. Classification results for all the methods using the Indian Pines dataset. (a) HybridSN. (b) A2S2KResNet. (c) S3NET. (d) DM-MRN. (e) CEGCN. (f) EMS-GCN. (g) WFCG. (h) DT. (i) RF. (j) FST-RF. (k) Semi-HCN. (l) GT.

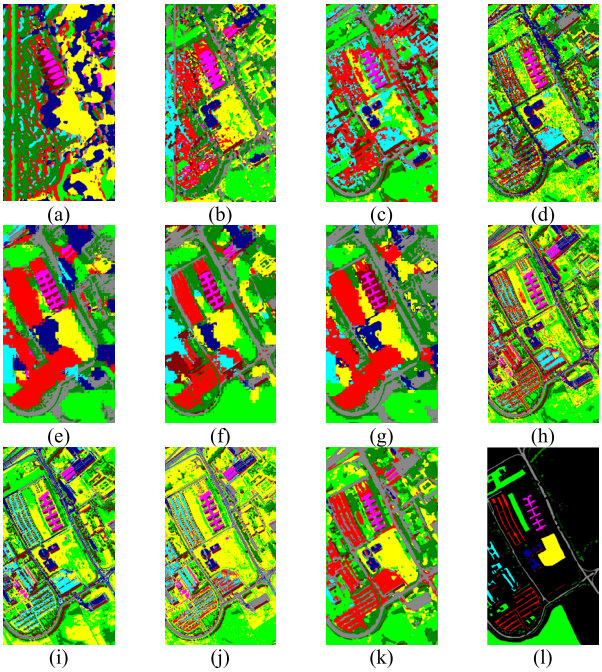


Fig. 10. Classification results for all the methods using the Pavia University dataset. (a) HybridSN. (b) A^2S^2KResNet. (c) S^3NET. (d) DM-MRN. (e) CEGCN. (f) EMS-GCN. (g) WFCG. (h) DT. (i) RF. (j) FST-RF. (k) Semi-HCN. (l) GT.

steps, with the average intraclass distance serving as the threshold. To ensure a fair comparison, we set the iteration to stop when the number of generated pseudo-labels exceeds 5000 for the Indian Pines and Pavia University datasets, and 10 000 for the WHU-Hi-LongKou dataset. In Table VIII, "A" represents the baseline, "B" indicates adaptive thresholding, and "C" represents ensemble learning-based filtering.

Table VIII shows the results of different module combinations on three datasets. There are three main modules in Semi-HCN, including the FST module, hierarchical multibranch network, and Hi-CE Loss, removing these three modules from Semi-HCN is the baseline mode. For the hierarchical multibranch network module, we decompose it into the multibranch structure and hierarchical labels to verify whether

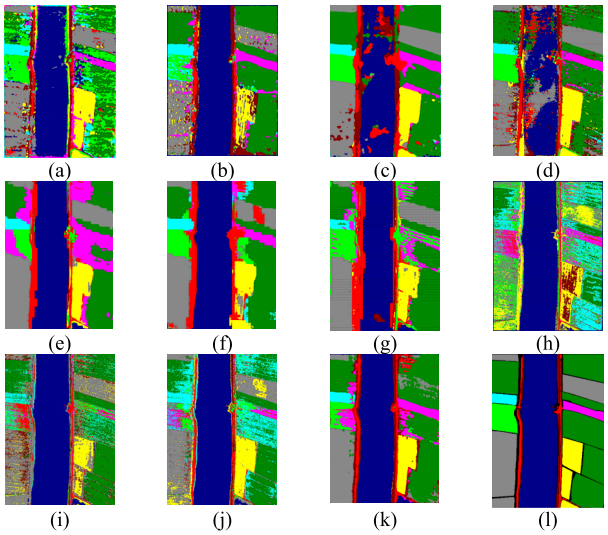


Fig. 11. Classification results for all the methods using the WHU-Hi-LongKou dataset. (a) HybridSN. (b) A2S2KResNet. (c) S3NET. (d) DM-MRN. (e) CEGCN. (f) EMS-GCN. (g) WFCG. (h) DT. (i) RF. (j) FST-RF. (k) Semi-HCN. (l) GT.

the effectiveness of the hierarchical multibranch network originates from better feature extraction or the application of hierarchical structures. The multibranch structure represents the constraint of each branch using the original labels, while hierarchical labels refer to the constraint of each branch using hierarchical labels based on the multibranch structure.

Since the FST module determines whether to stop iteration after a full round, the number of pseudo-labels generated in each iteration is not fixed under different strategies. This results in differences in the total number of pseudo-labels generated across strategies, although the overall quantity is similar. The results in Table VII show that the adaptive threshold and ensemble learning filtering strategies effectively improve the accuracy of pseudo-labels. In the Indian Pines dataset, adding these two strategies improved the accuracy of pseudo-labels by 8.31%. However, in the University datasets dataset, because the data of the same class are more concentrated and intraclass similarity is high, the baseline strategy already yields pseudo-labels with 100% accuracy, so the improvements from the two strategies are not evident. From the time perspective, since the adaptive threshold and ensemble learning filtering impose stricter requirements on pseudo-labels, fewer pseudo-labels are generated in each iteration. Therefore, more iterations are needed to generate the same number of pseudo-labels.

From Table VIII, it can be seen that the FST module provides the greatest improvement in accuracy. This is because the FST module can generate a large number of correct pseudo-labels, providing sufficient prior information for network training. After adding the FST module, the accuracy of the baseline improved by 21.51%, 15.79%, and 11.85%, respectively. For the hierarchical multibranch network, it can be observed that by applying hierarchical labels to constrain each branch, the accuracy improved by 1.11%, 0.58%, and 0.78%, respectively, demonstrating the effectiveness of the hierarchical structure. Overall, after using

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
NUMBER OF PSEUDO-LABELS, ACCURACY, COMPUTATION TIME, AND THE CORRESPONDING OA UNDER DIFFERENT MAXIMUM ITERATION

T	Indian Pines				Pavia University				WHU-Hi-LongKou			
	OA(%)	Number	Acc(%)	Time(s)	OA(%)	Number	Acc(%)	Time(s)	OA(%)	Number	Acc(%)	Time(s)
0	41.86	-	-	-	61.67	-	-	-	82.27	-	-	-
10	64.45	1058	99.55	0.73	63.78	787	100	0.38	90.25	925	100	0.53
20	67.69	1730	99.45	2.66	71.16	1703	100	1.47	90.67	2452	99.70	2.67
30	68.47	2598	99.55	4.45	74.57	2445	100	3.63	91.77	5191	99.85	12.78
40	70.45	3294	95.80	8.10	77.32	2952	100	6.79	92.06	8749	99.86	50.22
50	71.78	3659	94.63	11.99	79.17	3644	100	10.34	92.32	12639	99.87	148.07
60	72.11	3943	94.74	15.97	80.53	5013	100	17.05	92.59	16975	99.90	391.42
70	72.75	4311	92.30	21.43	81.67	6598	100	29.46	93.14	22383	99.92	874.04
80	72.80	4825	91.78	28.58	82.45	7575	100	48.84	93.41	28667	99.88	1773.24
90	72.95	5212	90.05	37.04	82.87	9016	100	75.88	93.23	35749	99.73	3355.45
100	72.89	5608	89.63	45.30	83.45	9878	100	114.38	92.79	42609	99.68	5847.11

the hierarchical multibranch structure, the accuracy improved by 1.83%, 1.50%, and 4.16%, respectively, as features were extracted hierarchically from coarse to fine granularity. After applying Hi-CE Loss, the accuracy improved by 3.00%, 1.19%, and 2.10%, respectively. The highest accuracy is achieved when all three modules are used simultaneously. Additionally, it was observed that even without the FST module, the use of a multibranch structure and Hi-CE Loss can still improve classification accuracy. This indicates that our approach can overcome the limitations of hierarchical classification in small sample problems.

2) Parameter Sensitivity and Pseudo-Label Analysis: In Semi-HCN, two parameters need to be manually set, including the maximum iterations T in the FST module and the hyperparameter α in the Hi-CE Loss. Therefore, this section will study the impact of these parameters on Semi-HCN performance and also analyze the pseudo-labels generated under different maximum iterations T.

a) Maximum iterations T: To explore how maximum iterations T affects the generation of pseudo-labels, thereby affecting classification accuracy, we analyzed the number of pseudo-labels, pseudo-label accuracy, running time, and corresponding OA of maximum iterations T within the range of (10, 20, 30, 40, 50, 60, 70, 80, 90, and 100).Table IX shows the specific experimental results. From the table, it can be seen that with the continuous increase of the maximum number of iterations, the number of pseudo-labels, the running time, and OA show an upward trend, especially the running time. The greater the maximum number of iterations, the greater the increase in the running time. This is because the pseudo-labels generated in the previous round will be put into the next round of iteration to participate in the calculation. So, with the increasing number of pseudo-labels, more running time is required. On the contrary, there will be a downward trend in the accuracy of pseudo-labels. This is a problem of error accumulation that is prone to occur in self-training methods, as erroneous pseudo-labels will generate more erroneous pseudo-labels when participating in subsequent calculations. Although the accuracy of pseudo-labels shows a downward trend, the pseudo-labels generated by our method still maintain a high accuracy. In the case of two labeled samples of each class, Semi-HCN can generate 5608 pseudo-labels with an accuracy rate of nearly 90% in the Indian Pines dataset, 42 609 pseudo-labels with an accuracy rate of nearly 100% in the WHU-Hi-LongKou dataset, and 9878 pseudo-labels with the accuracy rate of 100% in Pavia University dataset. This shows that the FST module can generate a large number of correct pseudo-labels in a short time when there are insufficient samples, which fully proves the effectiveness of our FST module.

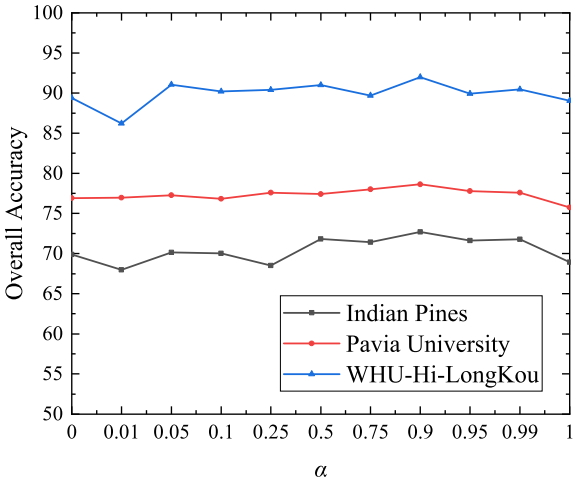


Fig. 12. OA versus different parameters α with the three datasets.

Finally, considering that too many iterations will consume a lot of calculation costs and time costs, we choose to set the maximum iterations $T = 50$ in this experiment. It is worth noting that when T is greater than 50, OA is still an upward trend, which means that our method can obtain better classification accuracy at the expense of computational efficiency.

b) Hyperparameter α: For the hyperparameter α, we set it in the range of (0, 0.01, 0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95, 0.99, and 1). Fig. 12 displays the classification results obtained using Semi-HCN with various parameters α. From the figure, it can be observed that the same trend is exhibited across the three datasets. The best OA is obtained when $\alpha = 0.9$, therefore we set $\alpha = 0.9$.

V. CONCLUSION

In this article, we validate the feasibility of using hierarchical structure information to assist deep learning in completing

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

HSI classification tasks through intuitive experiments. The results indicate that deep learning classification methods can achieve better classification performance with the assistance of hierarchical structure information. Therefore, we propose a Semi-HCN to achieve HSI hierarchical classification. This method first utilizes an FST module to obtain sufficient training samples, then extracts hierarchical features through a multibranch network, and finally applies hierarchical constraints to the output results using Hi-CE Loss. This method utilizes the hierarchical structure in the data to explore the relationships between different classes, achieving the goal of assisting HSI classification tasks.

The next focus of work is on how to construct appropriate hierarchical structures in HSIs. HSIs cannot directly construct hierarchical structures through semantic information, and manual design of hierarchical structures is required before training. However, from experimental results, it can be found that there are easily confused classes that are not under the same parent node, for example, C2 (meadows) and C6 (bare soil) in the Pavia University dataset. This indicates that our manual construction of hierarchical structures did not uncover the essential structural features in the data. Therefore, we will optimize the hierarchical construction method to achieve automatic and accurate mining of hierarchical structures in the data.

REFERENCES

[1] Y. Dong, X. Qin, X. Li, and L. Xu, "Multilevel spatial feature-based manifold metric learning for domain adaptation in remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5527414.

[2] D. Wang, J. Zhang, B. Du, L. Zhang, and D. Tao, "DCN-T: Dual context network with transformer for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 2536–2551, 2023.

[3] M. Wang, L. Gao, L. Ren, X. Sun, and J. Chanussot, "Hyperspectral simultaneous anomaly detection and denoising: Insights from integrative perspective," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 13966–13980, 2024.

[4] M. Wang, D. Hong, B. Zhang, L. Ren, J. Yao, and J. Chanussot, "Learning double subspace representation for joint hyperspectral anomaly detection and noise removal," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5507517.

[5] G. Sun, Z. Jiao, A. Zhang, F. Li, H. Fu, and Z. Li, "Hyperspectral image-based vegetation index (HSVI): A new vegetation index for urban ecological research," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102529.

[6] F. D. van der Meer et al., "Multi-and hyperspectral geologic remote sensing: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 14, no. 1, pp. 112–128, Feb. 2012.

[7] M. A. Calin, A. C. Calin, and D. N. Nicolae, "Application of airborne and spaceborne hyperspectral imaging techniques for atmospheric research: Past, present, and future," *Appl. Spectrosc. Rev.*, vol. 56, no. 4, pp. 289–323, Apr. 2021.

[8] J. Li, Z. Zhang, R. Song, Y. Li, and Q. Du, "SCFormer: Spectral coordinate transformer for cross-domain few-shot hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 33, pp. 840–855, 2024.

[9] M. A. Hearst, S. T. Dumais, E. Osuna, J. Platt, and B. Scholkopf, "Support vector machines," *IEEE Intell. Syst. Appl.*, vol. 13, no. 4, pp. 18–28, Aug. 1998.

[10] J. M. Keller, M. R. Gray, and J. A. Givens, "A fuzzy K-nearest neighbor algorithm," *IEEE Trans. Syst. Man, Cybern.*, vol. SMC-15, no. 4, pp. 580–585, Jul. 1985.

[11] M. Belgiu and L. Drăguţ, "Random forest in remote sensing: A review of applications and future directions," *ISPRS J. Photogramm. Remote Sens.*, vol. 114, pp. 24–31, Apr. 2016.

[12] Z. Wang, B. Du, and Y. Guo, "Domain adaptation with neural embedding matching," *IEEE Trans. neural Netw. Learn. Syst.*, vol. 31, no. 7, pp. 2387–2397, Jul. 2019.

[13] Y. Yuan, J. Lin, and Q. Wang, "Hyperspectral image classification via multitask joint sparse representation and stepwise MRF optimization," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 2966–2977, Dec. 2016.

[14] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021.

[15] S. Li, W. Song, L. Fang, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Deep learning for hyperspectral image classification: An overview," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6690–6709, Sep. 2019.

[16] J. Yang, B. Du, Y. Xu, and L. Zhang, "Can spectral information work while extracting spatial distribution—An online spectral information compensation network for HSI classification," *IEEE Trans. Image Process.*, vol. 32, pp. 2360–2373, 2023.

[17] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

[18] M. Schuster and K. K. Paliwal, "Bidirectional recurrent neural networks," *IEEE Trans. Signal Process.*, vol. 45, no. 11, pp. 2673–2681, Nov. 1997.

[19] J. Yang, B. Du, D. Wang, and L. Zhang, "ITER: Image-to-pixel representation for weakly supervised HSI classification," *IEEE Trans. Image Process.*, vol. 33, pp. 257–272, 2024.

[20] H. Lee and H. Kwon, "Going deeper with contextual CNN for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 26, no. 10, pp. 4843–4855, Oct. 2017.

[21] S. Yu, S. Jia, and C. Xu, "Convolutional neural networks for hyperspectral image classification," *Neurocomputing*, vol. 219, pp. 88–98, Jan. 2017.

[22] M. E. Paoletti, J. M. Haut, J. Plaza, and A. Plaza, "A new deep convolutional neural network for fast hyperspectral image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 120–147, Nov. 2018.

[23] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[24] Q. Zhu et al., "A spectral–spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 52, no. 11, pp. 11709–11723, Nov. 2022.

[25] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[26] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D-2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2019.

[27] H. Yu, Z. Xu, K. Zheng, D. Hong, H. Yang, and M. Song, "MSTNet: A multilevel spectral–spatial transformer network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5532513.

[28] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.

[29] A. J. Myles, R. N. Feudale, Y. Liu, N. A. Woody, and S. D. Brown, "An introduction to decision tree modeling," *J. Chemometrics*, vol. 18, no. 6, pp. 275–285, Jun. 2004.

[30] J. R. Quinlan, *C4. 5: Programs for Machine Learning*. Amsterdam, The Netherlands: Elsevier, 2014.

[31] L. Rutkowski, M. Jaworski, L. Pietruczuk, and P. Duda, "The CART decision tree for mining data streams," *Inf. Sci.*, vol. 266, pp. 1–15, May 2014.

[32] A. Gordon, "A review of hierarchical classification," *J. Roy. Stat. Soc., Ser. A*, vol. 150, no. 2, p. 119, Jan. 1987.

[33] D. Koller and M. Sahami, "Hierarchically classifying documents using very few words," in *Proc. ICML*, Jul. 1997, pp. 170–178.

[34] C. N. Silla and A. A. Freitas, "A survey of hierarchical classification across different application domains," *Data Min. Knowl. Discov.*, vol. 22, no. 1, pp. 31–72, 2011.

[35] D. Li, A. Cong, and S. Guo, "Sewer damage detection from imbalanced CCTV inspection data using deep convolutional neural networks with hierarchical classification," *Autom. Construct.*, vol. 101, pp. 199–208, May 2019.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.

[36] Y. Meng, J. Shen, C. Zhang, and J. Han, "Weakly-supervised hierarchical text classification," *Proc. AAAI Conf. Artif. Intell.*, vol. 33, no. 1, pp. 6826–6833, Jul. 2019.
[37] R. A. Stein, P. A. Jaques, and J. F. Valiati, "An analysis of hierarchical text classification using word embeddings," in *Proc. Inf. Sci.*, vol. 471, Jan. 2019, pp. 216–232.
[38] A. R. S. Parmezan, D. F. Silva, and G. E. A. P. A. Batista, "A combination of local approaches for hierarchical music genre classification," in *Proc. ISMIR*, Oct. 2020, pp. 740–747.
[39] K. He et al., "Synergistic learning of lung lobe segmentation and hierarchical multi-instance classification for automated severity assessment of COVID-19 in CT images," *Pattern Recognit.*, vol. 113, May 2021, Art. no. 107828.
[40] Y. Wang, Z. Wang, Q. Hu, Y. Zhou, and H. Su, "Hierarchical semantic risk minimization for large-scale classification," *IEEE Trans. Cybern.*, vol. 52, no. 9, pp. 9546–9558, Sep. 2022.
[41] R. M. Pereira, D. Bertolini, L. O. Teixeira, C. N. Silla, and Y. M. G. Costa, "COVID-19 identification in chest X-ray images on flat and hierarchical classification scenarios," *Comput. Methods Programs Biomed.*, vol. 194, Oct. 2020, Art. no. 105532.
[42] J. Chen and Y. Qian, "Hierarchical multilabel ship classification in remote sensing images using label relation graphs," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611513.
[43] M. R. Mohammadi and F. Rustaee, "Hierarchical classification of fine-art paintings using deep neural networks," *Iran J. Comput. Sci.*, vol. 4, no. 1, pp. 59–66, Mar. 2021.
[44] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.
[45] Z. Han et al., "Multisource collaborative domain generalization for cross-scene remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5535815.
[46] C. Pu, Y. Liu, S. Lin, X. Shi, Z. Li, and H. Huang, "Multimodal deep learning for semisupervised classification of hyperspectral and LiDAR data," *IEEE Trans. Big Data*, vol. 11, no. 2, pp. 821–834, Apr. 2025.
[47] B. Zoph et al., "Rethinking pre-training and self-training," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 3833–3845.
[48] L. Bruzzone, M. Chi, and M. Marconcini, "A novel transductive SVM for semisupervised classification of remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 11, pp. 3363–3373, Nov. 2006.
[49] X. Lu, J. Zhang, T. Li, and Y. Zhang, "A novel synergetic classification approach for hyperspectral and panchromatic images based on self-learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4917–4928, Aug. 2016.
[50] P. S. S. Aydav and S. Minz, "Classification of hyperspectral images using self-training and a pseudo validation set," *Remote Sens. Lett.*, vol. 9, no. 11, pp. 1109–1117, Nov. 2018.
[51] X. Gu, "A self-training hierarchical prototype-based approach for semi-supervised classification," *Inf. Sci.*, vol. 535, pp. 204–224, Oct. 2020.
[52] M. Lin, Q. Chen, and S. Yan, "Network in network," 2013, *arXiv:1312.4400*.
[53] Y. Dong, B. Du, L. Zhang, and L. Zhang, "Dimensionality reduction and classification of hyperspectral images using ensemble discriminative local metric learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2509–2524, May 2017.
[54] Y. Dong, Y. Jin, and S. Cheng, "Clustered multiple manifold metric learning for hyperspectral image dimensionality reduction and classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516813.
[55] Y. Zhong, X. Hu, C. Luo, X. Wang, J. Zhao, and L. Zhang, "WHU-hi: UAV-borne hyperspectral with high spatial resolution (H_2) benchmark datasets and classifier for precise crop identification based on deep convolutional neural network with CRF," *Remote Sens. Environ.*, vol. 250, Dec. 2020, Art. no. 112012.
[56] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.
[57] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.
[58] J. Zeng, Z. Xue, L. Zhang, Q. Lan, and M. Zhang, "Multistage relation network with dual-metric for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5510017.
[59] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2020.
[60] H. Zhang, J. Zou, and L. Zhang, "EMS-GCN: An end-to-end mixhop superpixel-based graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526116.
[61] Y. Dong, Q. Liu, B. Du, and L. Zhang, "Weighted feature fusion of convolutional neural network and graph attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 1559–1572, 2022.
[62] N. B. R and N. K. Rana, "A survey on decision tree algorithm for classification," *Int. J. Eng. Develop.*, vol. 2, no. 1, pp. 1–5, Mar. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 21:29:34 UTC from IEEE Xplore. Restrictions apply.